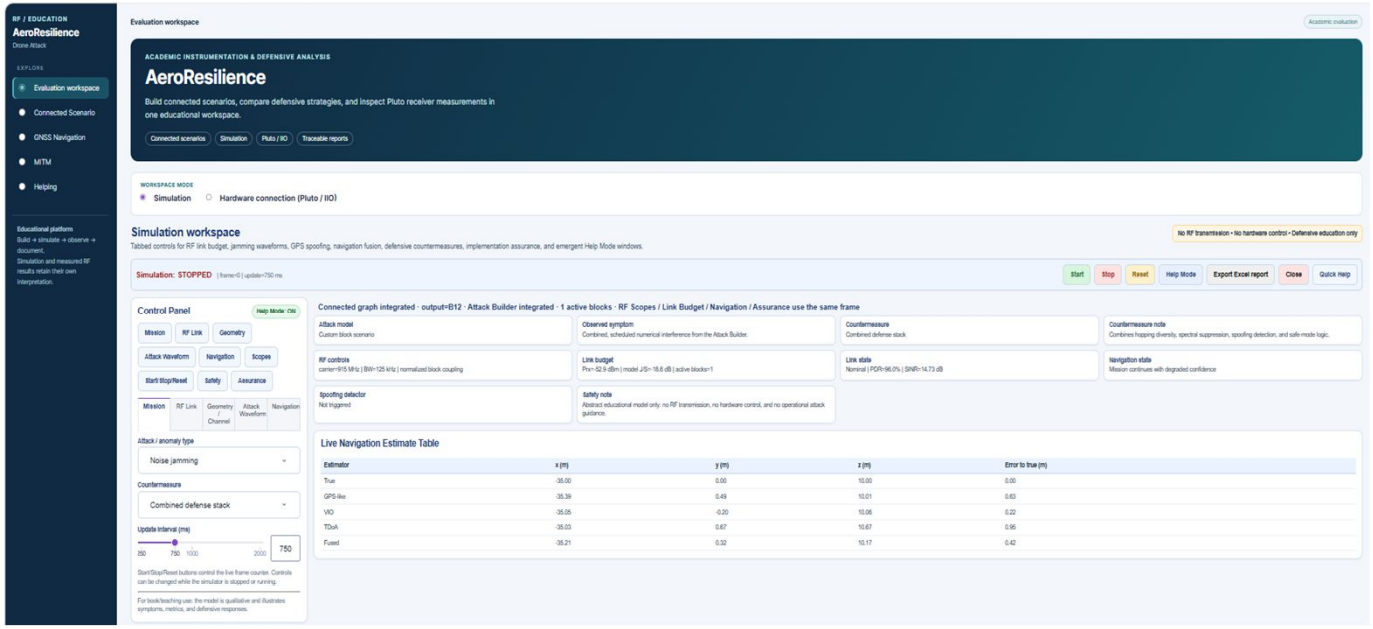
AeroResilience

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

ACADEMIC INSTRUMENTATION & DEFENSIVE ANALYSIS
AeroResilience

Build connected scenarios, compare defensive strategies, and inspect Pluto receiver measurements in one educational workspace.

AeroResilience



How to read this view

Choose the workspace mode

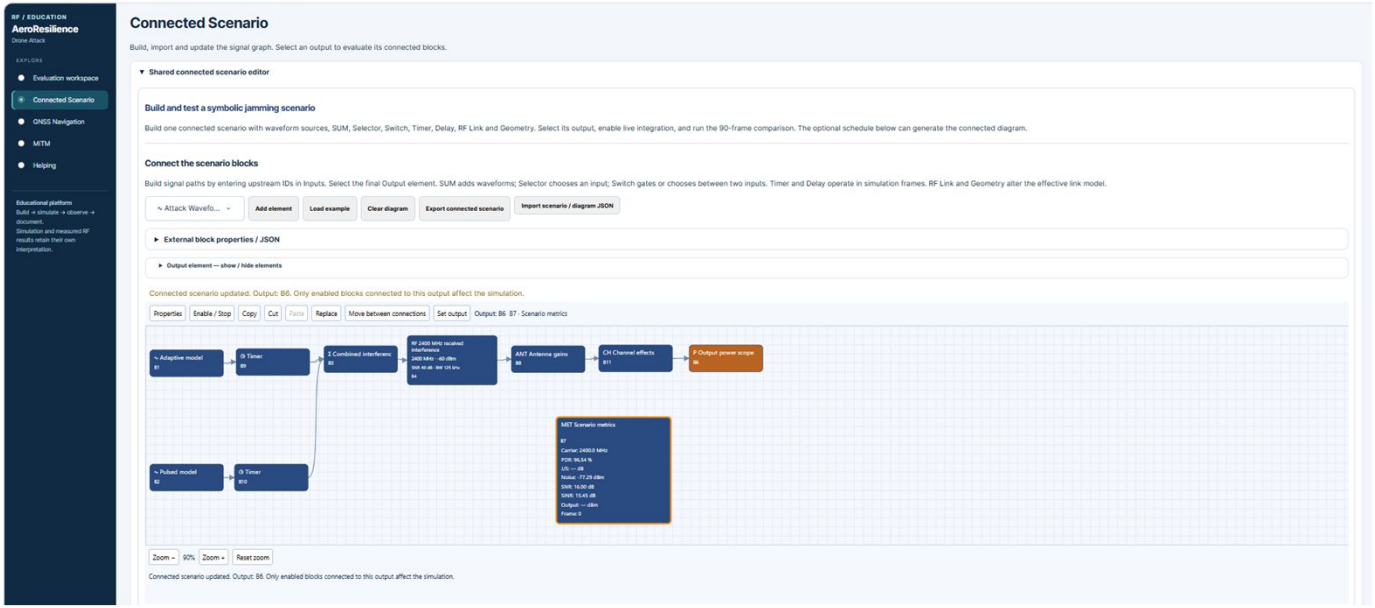
Use Simulation for the academic lab. The workspace status shows whether the run is stopped or active and which frame is displayed.

Read the scenario summary

Check the selected output, active blocks, modeled link condition and navigation status before comparing runs.

Compare position estimates

The table reports x, y and z in metres. Read Error to true as the distance between each estimate and the simulated reference.



How to read this view

Follow the signal graph

Read the arrows from source blocks toward the selected Output. Only enabled blocks connected to that output contribute to the result.

Check the selected output

Confirm the output ID and integration status before interpreting the plots. Use the same output in each comparison.

Save the scenario

Export the connected scenario JSON so another student can reload the same graph.



How to read this view

Branch frequencies

Read carrier frequency against simulation frame. A configured branch can appear here even when it is inactive or outside the selected receiver band.

Power measurement scopes

Compare modeled stage power and noise in dBm at the same frame. Use the stage labels to see where power changes along the connected path.



How to read this view

Connected output schedule

Read each row as the active frames for one connected block. Use the schedule to check whether an event overlaps the frame shown in the RF views.

Live link budget

Compare desired power, in-band interference and noise in dBm. Read SINR in dB and expected packet delivery as a modeled percentage.



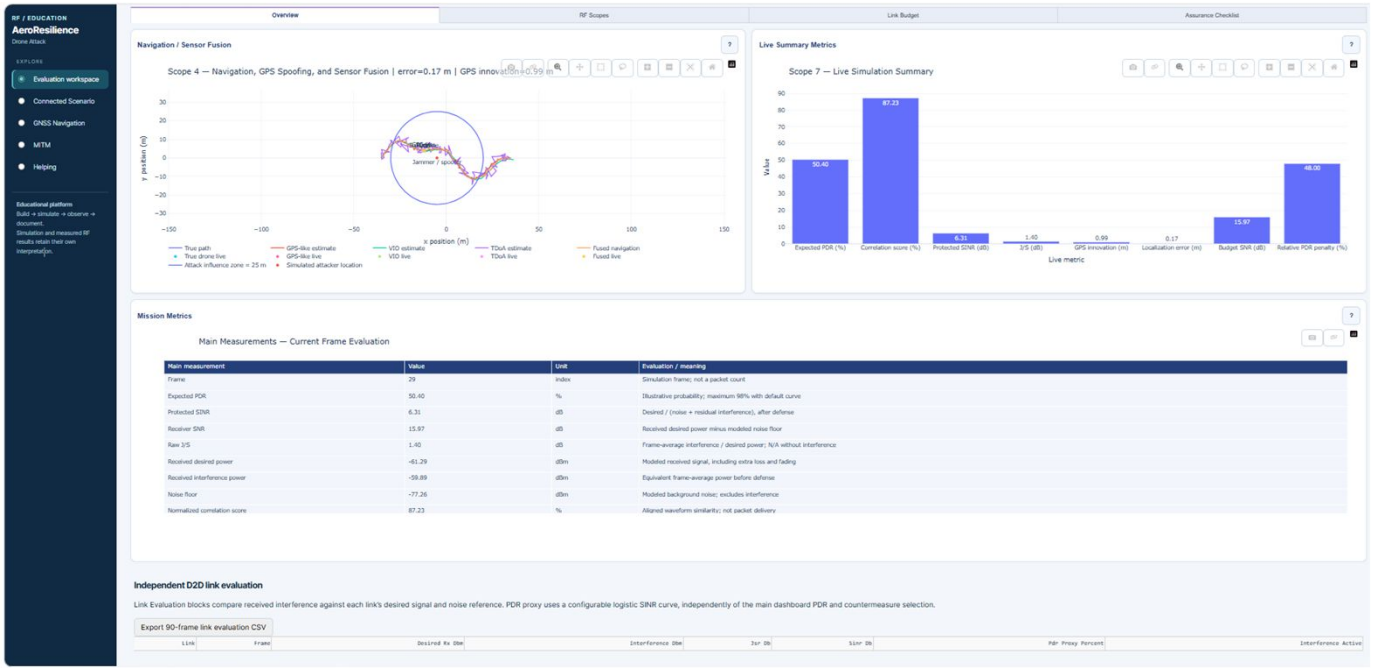
How to read this view

GNSS position error

Read synthetic position error against simulation frame. Compare the estimates before, during and after the modeled disturbance, using the same reference trajectory.

Navigation block evaluation

Use the frame table to align error changes with the simulated event and the navigation response. The plot and table describe synthetic navigation data.



How to read this view

Navigation and sensor fusion

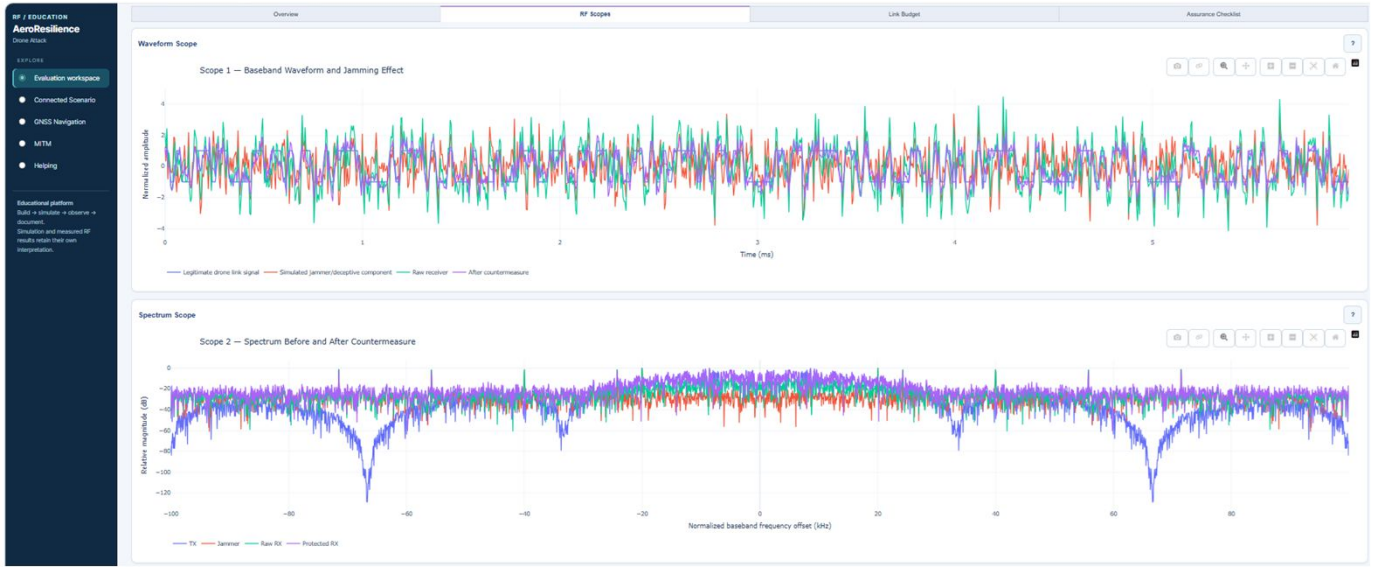
Compare the plotted estimates with the reference trajectory. Look for divergence, then check whether the fused estimate returns toward the reference.

Live simulation summary

Use the bars and measurements table to compare current-frame results. Check units before comparing received power, SINR, expected PDR and position error.

Independent D2D evaluation

Read each link separately. Its modeled delivery estimate can differ from the dashboard's main link estimate.



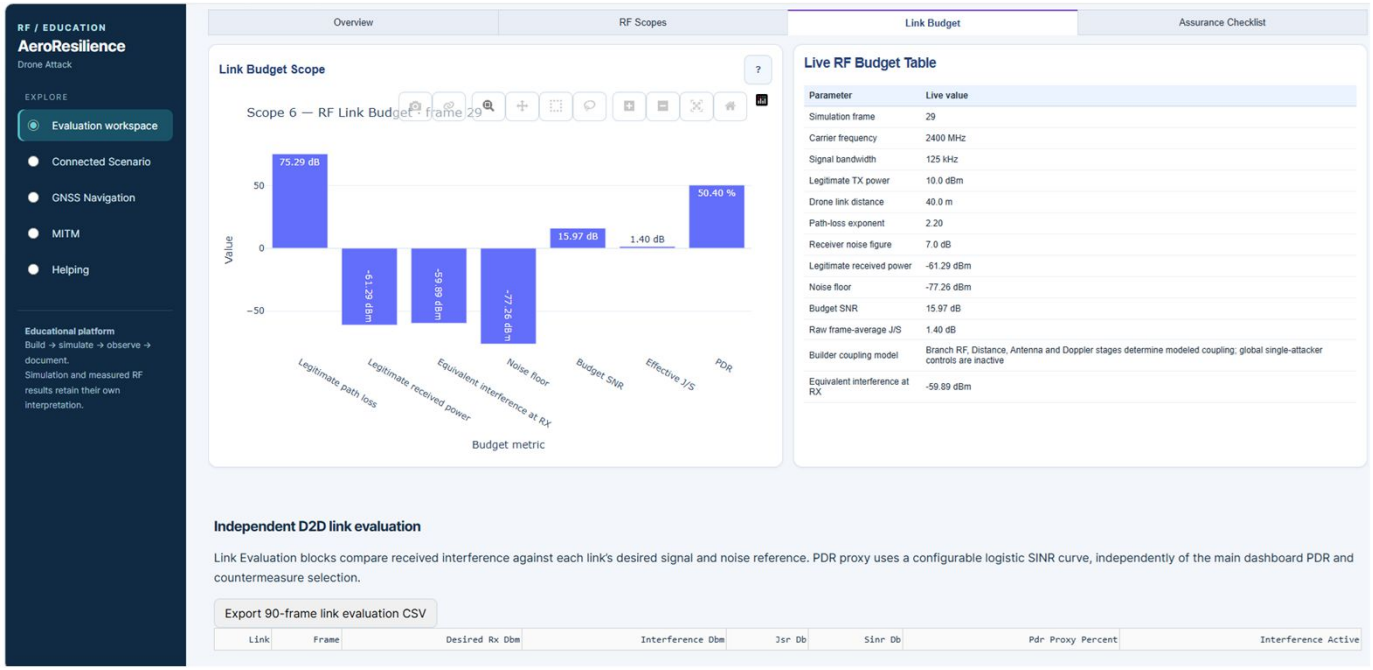
How to read this view

Waveform scope

Read baseband amplitude against time. Compare the traces to see when the modeled disturbance overlaps the desired signal.

Spectrum scope

Read the horizontal axis as baseband frequency offset. Compare the before and after traces in the same band to see where the modeled defense changes spectral energy.



How to read this view

RF link budget scope

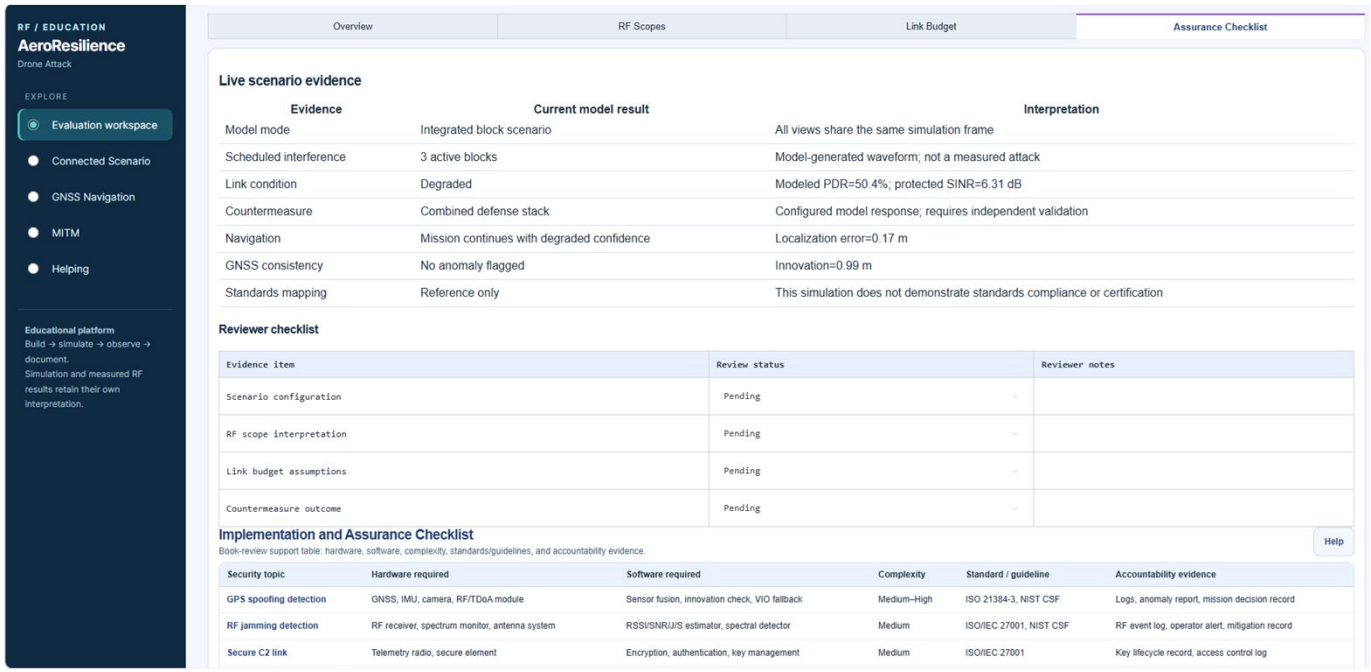
Compare received signal, interference and noise at the selected frame. Read power in dBm and ratios in dB. Expected PDR is a modeled percentage.

Live budget table

Check carrier, bandwidth, path loss and other assumptions behind the plot. Record the frame and receiver band when comparing cases.

Independent D2D evaluation

Compare the desired signal, interference and SINR for each link. Use the per-link delivery proxy to compare links under the same conditions.



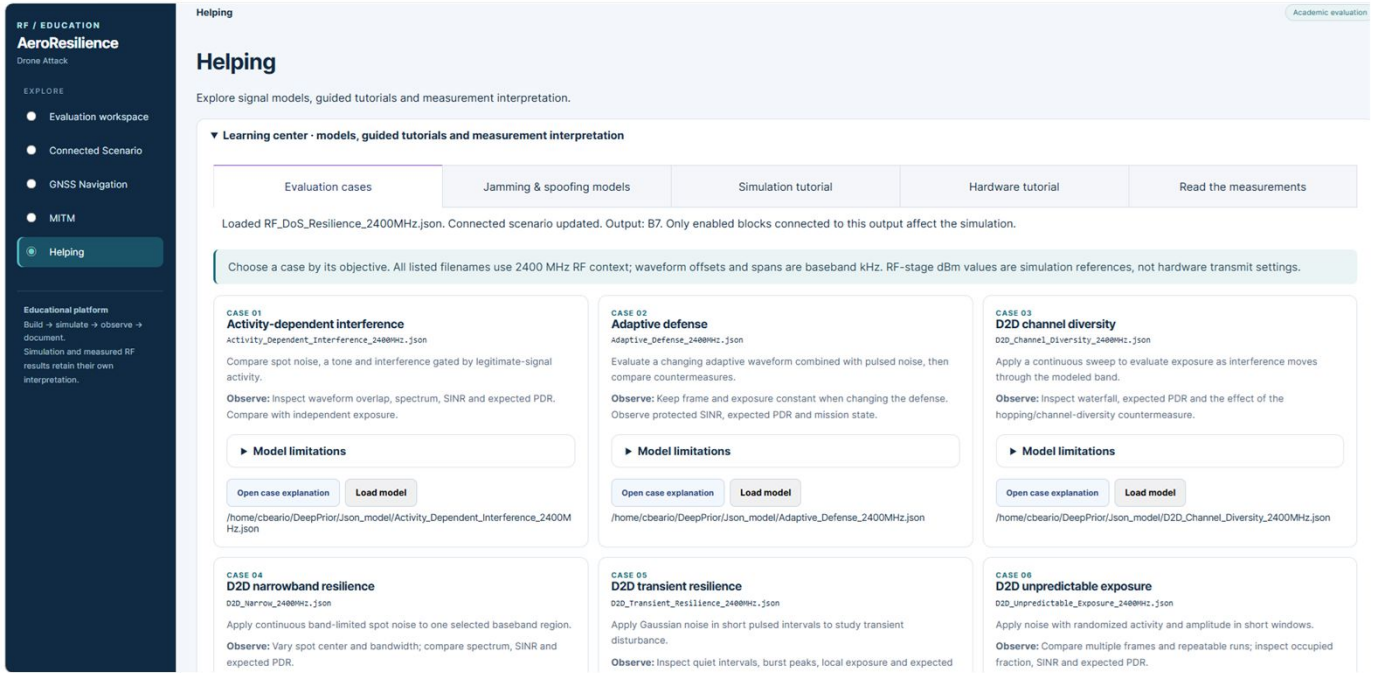
How to read this view

Read the evidence table

Connect each reported value to its interpretation. Record the frame and distinguish modeled link estimates from navigation results.

Use the reviewer checklist

Record which assumptions and results were reviewed. Treat standards references as pointers for further assessment.



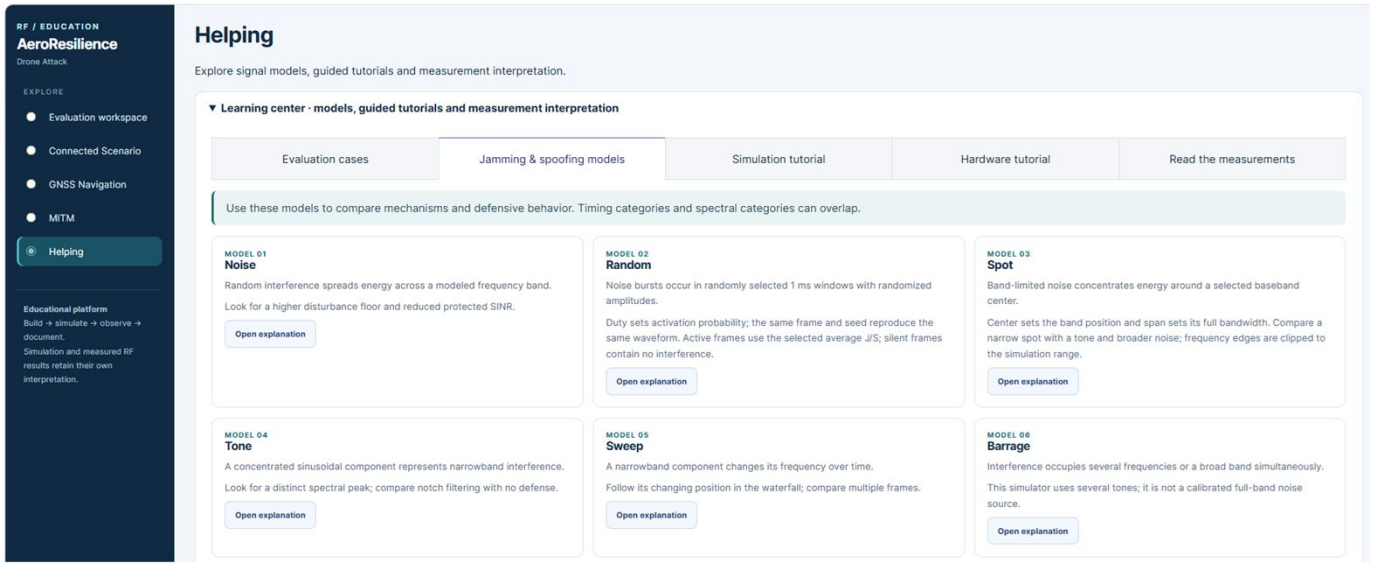
How to read this view

Choose by learning objective

Read what each evaluation case is designed to show and which scopes to inspect.

Check the loaded case

Confirm the scenario filename and selected output. Save the case name with your results so comparisons can be reproduced.



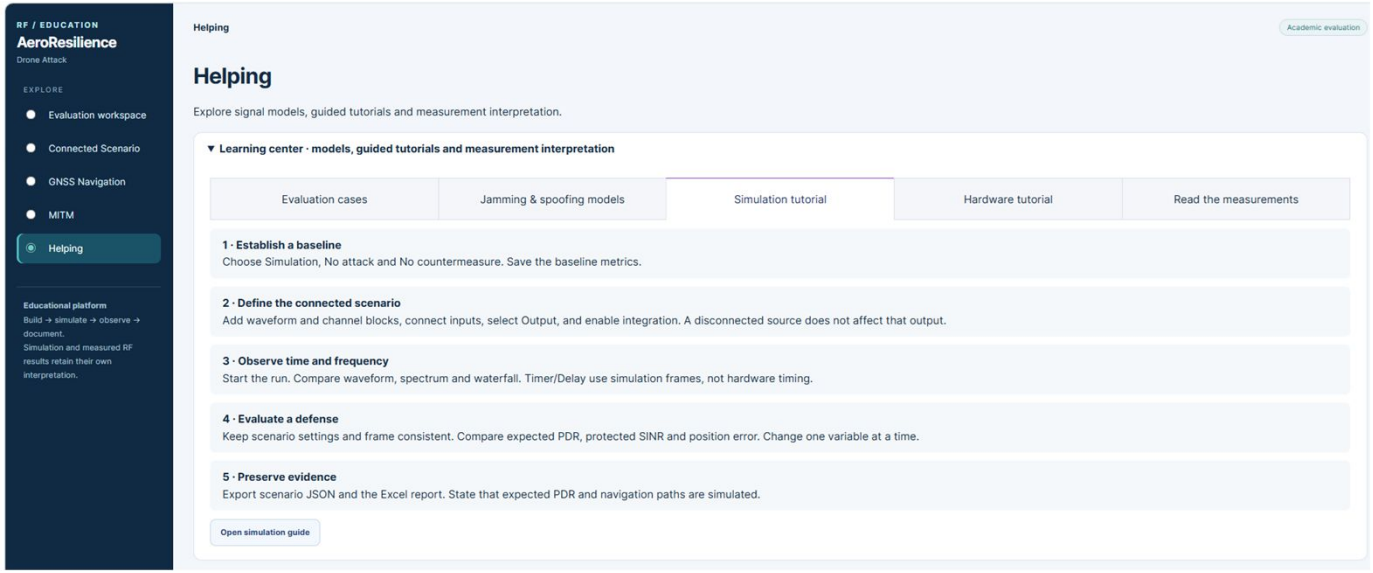
How to read this view

Compare modeled signatures

Use the model descriptions to distinguish broadband and narrowband effects in the simulated RF views.

Cross-check the plots

Compare the waveform, spectrum and modeled SINR for the same frame before interpreting a change.



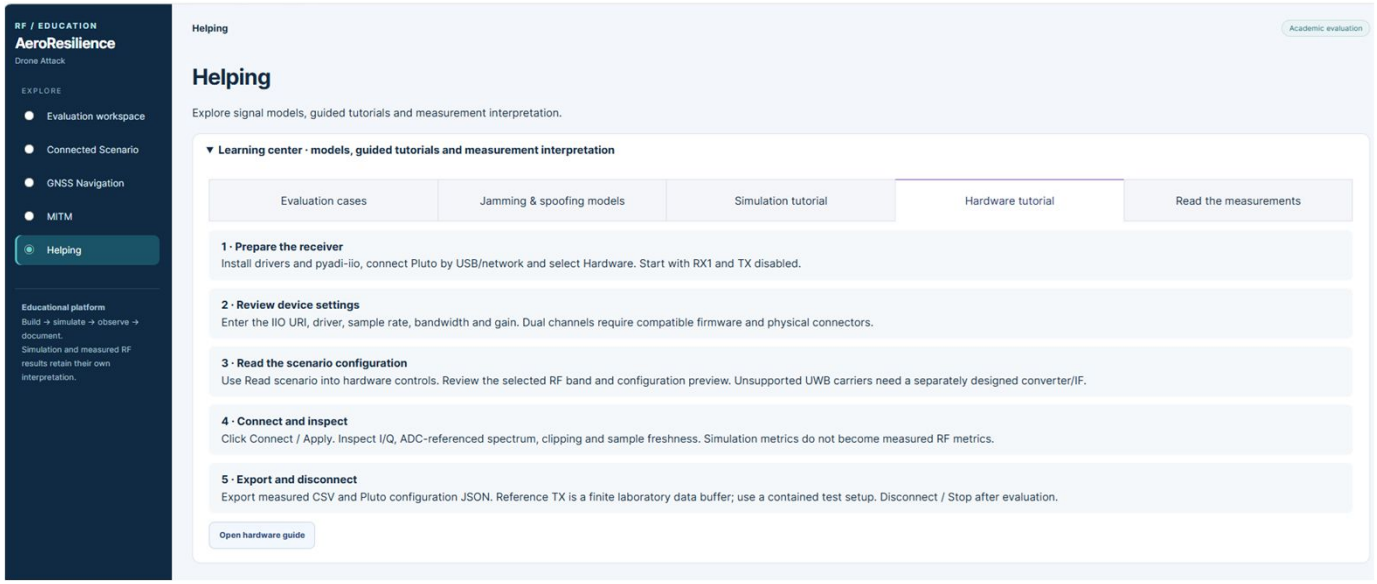
How to read this view

Start with a baseline

Save nominal link and navigation metrics. Use the same comparison frame and scenario settings when evaluating a defense.

Export the comparison

Save the scenario JSON and Excel report. Note which settings changed and identify reported values as simulation estimates.



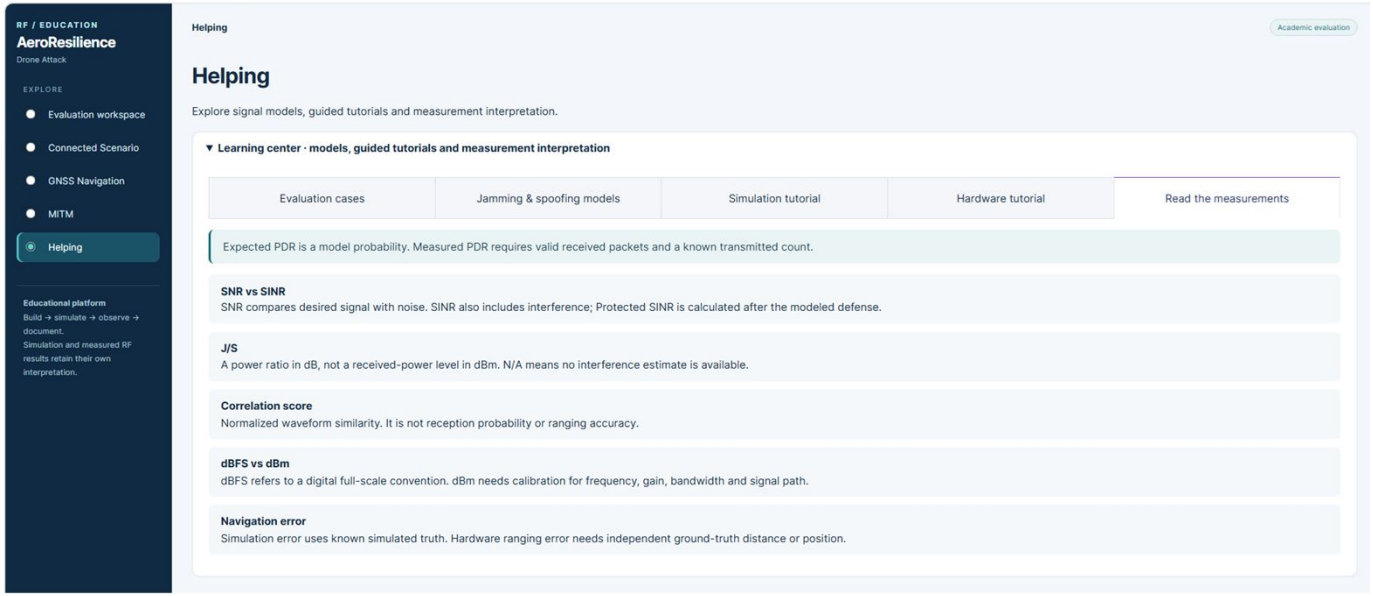
How to read this view

Read the receiver display

In receive-only mode, inspect I/Q samples, spectrum, clipping and sample freshness. Record the acquisition settings with each capture.

Interpret the reference scale

Treat ADC-referenced spectrum values as relative readings unless the receiver chain has been calibrated for absolute power.



How to read this view

Interpret PDR

Expected packet delivery ratio is a model estimate. Measuring PDR requires both a valid received packet count and a known transmitted count.

Distinguish SNR and SINR

SNR compares desired signal with noise. SINR also includes interference. Both are ratios in dB. Power levels are reported in dBm.

Academic simulation lab

How does a defense change link quality and navigation under the same simulated disruption?

1. Save a baseline

In Helping → Simulation tutorial, select Simulation, No attack, and No countermeasure. Save the baseline metrics.

2. Compare two runs

Use one supplied evaluation case. Confirm the selected Output is integrated. Run it with the defense off and on. Keep the scenario and comparison frame fixed.

3. Explain the result

Compare expected PDR, protected SINR, and position error. Report any trade-off between link quality and navigation.

Submit the evidence

Export the scenario JSON and Excel report. Submit a three-run comparison table and label the results as simulated. Report N/A where the tool provides no estimate.